

Energy efficiency

ENERGY IS LOST AS HEAT BY ALL MACHINES AND PROCESSES.

When a machine or activity is designed or planned, it should be as efficient as possible. This means that as much as possible of the energy output should be used for work.

Lost energy

At a subatomic scale, some processes occur with 100 percent transfer of energy from one form to another, but on a larger scale, this never happens. Some energy is always turned into heat, which is usually unwanted. Other types of unwanted energy may also be produced: many machines make a lot of noise, which is a wasteful and unpleasant form of acoustic energy.

▽ Energy conversion

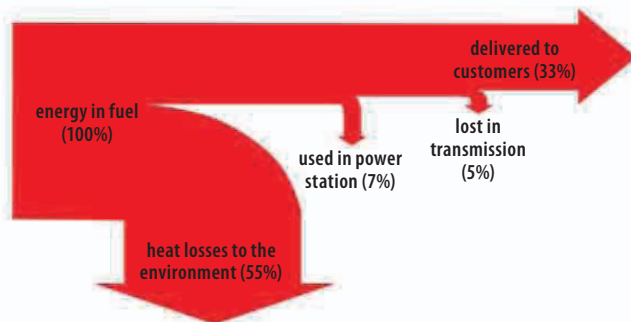
Every type of energy conversion has a maximum possible efficiency. Some processes are wasteful, while others convert a very high proportion of one form of energy into another.

SEE ALSO

- ◀ 170–171 Energy
- ◀ 188–189 Heat transfer
- ◀ 216–217 Transformers

▽ Energy loss

Only a third of the energy consumed by thermal power plants reaches customers as electricity.



CONVERSION EFFICIENCIES

Energy process	Conversion taking place	Maximum possible efficiency
photosynthesis	radiant energy from the Sun to chemical energy in the plant	6%
solar cell	radiant energy from the Sun to electrical energy, often produced by silicon crystals	28%
muscle	chemical energy from chemicals in the blood to kinetic energy as the muscle contracts	30%
coal-fired power station	chemical energy from coal to electrical energy from turbines	40%
internal combustion	chemical energy from gasoline or diesel to kinetic energy used to make vehicle move	50%
wind turbine	kinetic energy of the wind to electrical energy from a generator	60%
electric heater	electrical energy to thermal energy, produced by electrical resistance of the element	100%

REAL WORLD

Fiberoptic cable

Until a few decades ago, telephone and computer signals were usually sent in the form of electric currents that traveled down copper wires. Although copper conducts electricity very well, some of the electrical energy is lost in the form of heat. Now, optical fibers have replaced many copper wires. In an optical fiber, signals travel in the form of light, and only a tiny amount of the light energy is changed to heat, making it a far more efficient system.



Many **household appliances** are not energy efficient—the only 100 percent efficient device is the **electric heater**. However, they can be very expensive to use.